SA01 ASSESS VENTILATION | O (MAX: 1 PT)

Intent: Minimize indoor air quality issues through the provision of adequate ventilation.

Summary: This feature requires projects to assess their ability to bring in fresh air from the outside through mechanical and/or natural means in order to dilute human- and product-generated air pollutants.

Issue: Poorly ventilated spaces contribute to symptoms — such as headache, fatigue, dizziness, nausea, cough, sneezing, shortness of breath and eye, nose, throat and skin irritation — collectively called sick building syndrome (SBS).[203,204] Poor ventilation is also linked to increased rates of absences in employees, higher operational costs for businesses and decreased productivity in students.[205,206] One U.S.-based study reported that the sick leave attributable to insufficient provision of fresh air in buildings is estimated to be 35% of total absenteeism.[207] Therefore, the economic costs of SBS in under-ventilated buildings are significant and far exceed the energy-related cost savings.[208–210]

Solutions: Many indoor and outdoor sources of air pollution emit particulate matter and volatile organic compounds (VOCs) that can cause discomfort and trigger asthma and eye, nose and throat irritation. In order to maintain healthy indoor environments for building users, it is necessary to provide sufficient ventilation required to maintain acceptable air quality.[211,212] Increasing ventilation rates is also a recommended strategy to mitigate the transmission of COVID-19 and other airborne contagious diseases.[213,214] In addition to proper HVAC system design, mechanically ventilated projects need to perform regular system maintenance as inadequate maintenance is associated with reduced ventilation performance and poorer indoor air quality and thermal conditions.[215]

Part 1 Part 1 (Max: 1 Pt)

For All Spaces:

A qualified engineer provides the project with an assessment of the following:

a. The extent to which the current mechanical system can operate without recirculating air

b. How and if any of the potential HVAC system modifications would affect the following:

    1. Energy consumption.
    2. The ability to manage thermal comfort conditions (e.g., higher ventilation leading to draft, recirculation elimination straining conditioning capacity).
    3. Maintenance processes.

c. The highest supply rate of outdoor air the current mechanical system can provide.

d. Potential modifications to system controls to increase supply of outdoor air (e.g., ventilating for longer hours, changing the setpoint for demand-controlled ventilation systems).